
```

clc; clear;

% Input data format: [From To R X]
linedata = [
    1  2    0.07    0.15;
    1  3    0.06    0.10;
    1  4    0.08    0.25;
    2  4    0.04    0.10;
    3  4    0.04    0.20;
];

n = 4;
slack_bus = 1;
P_inj = [0; -0.4; -0.5; -0.7]; % P injections
V = [1.05; 1.0; 1.0; 1.0];      % Voltage magnitudes

fprintf('DC POWER FLOW ANALYSIS (Optimized)\n\n');

% Calling Y-Bus Function
Ybus = formYbus(linedata, n);
fprintf('Y_bus Matrix:\n'); disp(Ybus);

% Form B Matrices & Solve Angle
% B' = -imag(Ybus) for DC approximation
B = -imag(Ybus);

% Reduce B matrix (remove slack bus row/col)
non_slack = setdiff(1:n, slack_bus);
B_reduced = B(non_slack, non_slack);
P_known = P_inj(non_slack);

% Solve: P = B * delta --> delta = B \ P
delta_calc = B_reduced \ P_known;

% Reconstruct full delta vector
delta = zeros(n, 1);
delta(non_slack) = delta_calc;

% Calculate complex voltage V = |V| /_ delta
V_comp = V .* exp(1j * delta);

% Calculate S = V * conj(I) = V * conj(Y * V)
S_inj = V_comp .* conj(Ybus * V_comp);
P_calc = real(S_inj); % Actual P (includes losses)
Q_calc = imag(S_inj); % Reactive Power

% Results
fprintf('\nBUS RESULTS:\n');
fprintf('Bus | Angle(deg) | V (pu) | P_inj(pu) | Q_calc(pu)\n');
fprintf('----|-----|-----|-----|-----\n');
for i = 1:n
    fprintf(' %d | %9.4f | %4.4f | %8.4f | %9.4f\n', ...

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        i, rad2deg(delta(i)), V(i), P_inj(i), Q_calc(i));
    end

    fprintf('\nLINE FLOW RESULTS:\n');
    fprintf('From-To | P_flow (pu) | Angle Diff(deg)\n');
    fprintf('-----|-----|-----\n');

    for k = 1:size(linedata, 1)
        fb = linedata(k, 1);
        tb = linedata(k, 2);

        % Retrieve Yij magnitude from Ybus (off-diagonal is -yij)
        Y_mag = abs(Ybus(fb, tb));

        ang_diff = delta(fb) - delta(tb);

        % Power Flow Equation: P = Vi*Vj*Yij*sin(delta_i - delta_j)
        P_flow = V(fb) * V(tb) * Y_mag * sin(ang_diff);

        fprintf(' %d->%d | %8.4f | %4f\n', ...
            fb, tb, P_flow, rad2deg(ang_diff));
    end

    % Calculate Slack Bus Generation (Sum of flows leaving bus 1)
    (Approximated by summing P_calc indices - simplified check)
    fprintf('\nTotal P Mismatch/Loss: %4f pu\n', sum(real(S_inj)) -
sum(P_inj));

DC POWER FLOW ANALYSIS (Optimized)

Y_bus Matrix:
    8.1276 -16.4558i  -2.5547 + 5.4745i  -4.4118 + 7.3529i  -1.1611 + 3.6284i
   -2.5547 + 5.4745i   6.0030 -14.0951i   0.0000 + 0.0000i  -3.4483 + 8.6207i
   -4.4118 + 7.3529i   0.0000 + 0.0000i   5.3733 -12.1606i  -0.9615 + 4.8077i
   -1.1611 + 3.6284i  -3.4483 + 8.6207i  -0.9615 + 4.8077i   5.5709 -17.0568i

BUS RESULTS:
Bus | Angle(deg) | V (pu) | P_inj(pu) | Q_calc(pu)
----|-----|-----|-----|-----
1 | 0.0000 | 1.0500 | 0.0000 | 0.1391
2 | -5.6761 | 1.0000 | -0.4000 | -0.0360
3 | -4.9738 | 1.0000 | -0.5000 | 0.0374
4 | -6.6221 | 1.0000 | -0.7000 | 0.0723

LINE FLOW RESULTS:
From-To | P_flow (pu) | Angle Diff(deg)
-----|-----|-----
1->2 | 0.6274 | 5.6761
1->3 | 0.7806 | 4.9738
1->4 | 0.4613 | 6.6221
2->4 | 0.1533 | 0.9460
3->4 | 0.1410 | 1.6482

```

Total P Mismatch/Loss: 1.6995 pu

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